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From six brands to one: choosing a 50 kW hybrid inverter and battery for off-grid

Technical note · Inverter and high-voltage battery comparison · Rev.01, 7 November 2026

Summary

We started with six brands of 50 kW three-phase hybrid inverter, checked every figure against the manufacturers' datasheets, and removed the brands that do not fit an off-grid system. We then added each brand's own high-voltage battery in the 60–100 kWh range. The result: **Solis S6-EH3P50K-H with the Solis FlexCore-ID battery** is the best overall choice for a 50 kW off-grid site, with one sizing rule: 5 packs (100 kWh) when night loads stay below about 45 kW, 6 packs (about 120 kWh) when the battery alone must carry the full 50 kW.

1. Why we made this comparison

Many of the sites Solar ESS designs for are far from a reliable grid: homestays, resorts, farms and small factories in the hills. For these sites the inverter and the battery are the heart of the system. They must run the loads, refill the battery from the sun on the same day, start pumps and compressors, and hand over to a generator when the sun is not enough.

We wanted one clear answer for a 50 kW three-phase off-grid system, based on published datasheets rather than marketing claims.

2. How we worked

- We began with an AI-generated search summary that compared Deye and Solis, later extended to six brands.
- Every figure was checked against the manufacturer's datasheet, product page or installation manual.
- Where a figure could not be found in an official document, we marked it “not confirmed” instead of guessing.
- Brands were removed only for a clear technical reason: no suitable product, or key data that could not be verified.
- We judged the systems on what matters off-grid: PV oversizing, battery current and ports, surge capacity, generator support, switch-over time, battery energy, battery power and battery life.

3. Step 1 — Six brands, and what the datasheets really say

The first summary looked convincing, but several of its figures were wrong or could not be found in any official document:

Brand	What the summary claimed	What we found
Deye	3 or 4 MPPT; 55 kVA; 1.5× surge for 10 s; touchscreen	The 50 kW model has 4 MPPT (3 MPPT belongs to the smaller models). 55 kVA, the 10 s surge and the touchscreen were not confirmed.
Solis	97.8% efficiency; 7-inch “Zeitler” industrial screen	A 7-inch LCD with Bluetooth, no brand name. 97.8% was not confirmed. 140 A, 1.6× for 2 s, IP66 and 73 kg were all correct.
GoodWe	100 A as 50 A + 50 A on two ports	One battery input at 100 A. 98.1%, 200–800 V, IP66 and 65 kg were correct.
Growatt	7 to 10 MPPT; battery 600–1000 V	The 50 kW model has 7 MPPT (10 is the 100 kW model). Recommended battery voltage 768 V. It is a 153 kg floor cabinet.
SAJ	91.3 A × 2; 55 kVA; 6 MPPT	None of these figures could be found on a datasheet.

Brand	What the summary claimed	What we found
Sungrow	SG50CX-P2 with ST50CP / ST125CP storage blocks	SG50CX-P2 is a grid-tied string inverter. The ST50CP / ST125CP names were not found. Sungrow hybrids stop at 25 kW.

> Lesson: an AI summary is a useful starting list, but every number must be checked against the manufacturer's own documents before it goes into a design or a proposal.

4. Step 2 — Removing Sungrow

Sungrow has no single 50 kW hybrid inverter. Its hybrid range goes from 3 kW to 25 kW, and its commercial storage is built from separate power conversion systems and storage cabinets such as PowerStack, or the new 125 kW SH125CX. At 50 kW the only Sungrow hybrid route is two 25 kW units in parallel, which is a different kind of system. Sungrow was removed from the comparison.

5. Step 3 — Removing SAJ

For SAJ we could not confirm the battery current, the apparent power or the MPPT count of the 50 kW hybrid on any datasheet. Its 50 kW all-in-one cabinet (CHS2, 57.3 or 100.3 kWh, 150 A) is rated IP55. Solar ESS does not recommend equipment whose key figures cannot be verified, so SAJ was removed.

6. Step 4 — Four inverters compared for off-grid

Item	Solis	Deye	GoodWe	Growatt
Model	S6-EH3P50K-H	SUN-50K-SG01HP3-EU-BM4	GW50K-ET	WIT 50K-HU
MPPT	4	4	4	7
Battery ports / current	2 / 140 A (70+70)	2 / 100 A (50+50)	1 / 100 A	not confirmed
Battery voltage	150–800 V	160–800 V	200–800 V	768 V recommended
PV oversizing (max PV)	2.0× (100 kW)	~1.3× (65 kW)	1.5× (75 kW)	2.0×
Off-grid surge	1.6× for 2 s	75 kW peak	not confirmed	110% continuous
Switch-over time	< 10 ms	< 10 ms	< 10 ms with STS box	20 ms
Generator	Yes, 10–50 kVA	Yes	not confirmed	Yes, remote control
Parallel units	Up to 10	Up to 10	not confirmed	3 off-grid
Max efficiency	not confirmed	97.6%	98.1%	98.2%
IP rating / weight	IP66 / 73 kg	IP65 / ~80 kg	IP66 / 65 kg	153 kg floor cabinet

For off-grid, the sun has to run the loads and refill the battery on the same day, so PV oversizing and battery current matter most. Solis leads on both: twice its rated power in PV and 140 A across two independent battery ports. Deye has the strongest peak power for motor starts but takes the least PV. GoodWe is efficient and light, but has one battery port and needs an extra STS box for UPS-level switching. Growatt has generator control and black start, but it is a heavy floor cabinet with a three-unit limit in off-grid parallel.

7. Step 5 — Adding each brand's own battery (60–100 kWh)

At first Solis looked weak here, as it had no proven battery of its own in this size. That changed with the Solis FlexCore-ID, a stackable high-voltage battery of 20 kWh packs, 60–261 kWh per cluster. With it, every brand in the comparison now offers a matched inverter and battery from one maker.

Item	Solis FlexCore-ID	Deye BOS-G Pro	Growatt APX	GoodWe 100 kWh
Module	**20.09 kWh pack, 64 V**	**5.12 kWh, 51.2 V**	**14.33 kWh, 80 V**	**One cabinet, 672 V**
Sizes in range	60 / 80 / 100 kWh (3–5 packs)	61.4–87 kWh (12–17 modules); 100 kWh = 2 clusters	71.7 / 86 / 100.4 kWh (5–7 modules)	~100 kWh (Lynx C starts at 101 kWh)
Usable energy	100% DoD	90% DoD (87 → 78.3 kWh)	~90% (100.4 → 90.3 kWh)	100% DoD
Cycle life	≥ 8,000 at 100% DoD	≥ 6,000 at 90% DoD	not confirmed	Warranty: 70% at 10 years / 4,000 cycles (Lynx C)
Battery current	157 A rated	100 A (125 A peak, 2 min)	125 A	90 A charge / 120 A discharge
Enclosure	IP20, indoor	IP20, rack	IP66, outdoor	Lynx C: IP21

8. Step 6 — A finding: battery voltage limits battery power

Battery power is voltage × current. A FlexCore-ID pack is 64 V, so the battery voltage rises with the number of packs, while the Solis inverter takes at most 140 A. With 5 packs (about 320 V) the battery can give about 45 kW, and less when it is nearly empty. Our first kit assumed 100 kWh would run the full 50 kW for two hours; that was corrected. To carry the full 50 kW from the battery alone, 6 packs (about 384 V, about 54 kW) are needed.

Battery size	Solis + FlexCore-ID	Deye + BOS-G Pro	Growatt + APX	GoodWe + 100 kWh
~60 kWh	3 packs → ~27 kW	12 modules → 50 kW	—	—
~70–80 kWh	4 packs → ~36 kW	16 modules → 50 kW	5 modules → 35 kW	—
~90–100 kWh	5 packs → ~45 kW	17 modules (87 kWh) → 50 kW	7 modules → 50 kW	50 kW
~120 kWh	6 packs → 50 kW	2 clusters → 50 kW	8 modules → 50 kW	—

Figures are battery output with each brand's own 50 kW inverter, at nominal voltage. During the day the PV array adds its power on top, so at most off-grid homestays and resorts the night load, not the daytime peak, decides the battery size.

> Lifetime energy, as a guide: at 100 kWh, Solis gives about 800 MWh over its life (100 kWh × 8,000 cycles) and Deye about 470 MWh (78 kWh usable × 6,000 cycles), about 1.7 times more for Solis. These are manufacturer ratings and are indicative only.

9. Final ranking for a 50 kW off-grid system

Rank	System	Why
1	**Solis S6-EH3P50K-H + FlexCore-ID**	**One maker for inverter, battery, BMS and monitoring. 2× PV oversizing, 140 A on two ports, built-in generator mode, 100% usable energy and the longest battery life. Needs an indoor battery room.**
2	Deye 50K + BOS-G Pro	Full 50 kW from the battery at every size from 61 kWh, and 75 kW peak for motor starts. But only ~1.3× PV oversizing, 90% usable energy and a shorter rated life.
3	Growatt WIT 50K-HU + APX	The only outdoor-rated battery (IP66), black start and generator control. A heavy floor cabinet, 90% usable energy and a three-unit off-grid parallel limit.
4	GoodWe GW50K-ET + 100 kWh	Strong battery power and high efficiency, but one battery port and an STS box needed for UPS-level switching. Better suited to on-grid with backup.

10. Recommended 50 kW off-grid kit

Item	Specification
Inverter	Solis S6-EH3P50K-H, 50 kW three-phase hybrid
Battery, option A	FlexCore-ID, 5 packs ≈ 100 kWh, about 45 kW from the battery: for night loads below about 45 kW
Battery, option B	FlexCore-ID, 6 packs ≈ 120 kWh, full 50 kW from the battery
PV array	About 100 kWp, using the 2× oversizing
Backup source	Diesel generator 10–50 kVA on the generator port
Motor loads	Soft starters or VFDs on pumps, compressors and cold rooms (inverter surge 1.6× for 2 s)
Battery room	Dry and shaded (FlexCore-ID is IP20); FlexCore-OD, 48–253 kWh, for outdoor sites

11. Notes and limits

- Specifications are from manufacturer datasheets, product pages and manuals as of November 2026 and may change.
- Items marked “not confirmed” were not found in an official document and must be checked before use in a design.
- Some cycle-life figures are cell ratings rather than system ratings; lifetime energy is indicative only.
- Battery warranty terms in Thailand are confirmed with our supplier at quotation.
- This note compares technical specifications only. It contains no prices.

Solar ESS Working Group